

Data Science

EDA for Modeling

Exploratory Data Analysis



GENERAL ASSEMBLY

Learning objectives



1

Summarize the reasons and ways to explore data for the purpose of modeling and machine learning.

2

Define a random variable in the context of statistics and data.

3

Explain how identifying a variable's distribution helps inform our modeling approach.

4

Interpret correlation as a measure of the strength of the linear relationship between variables.

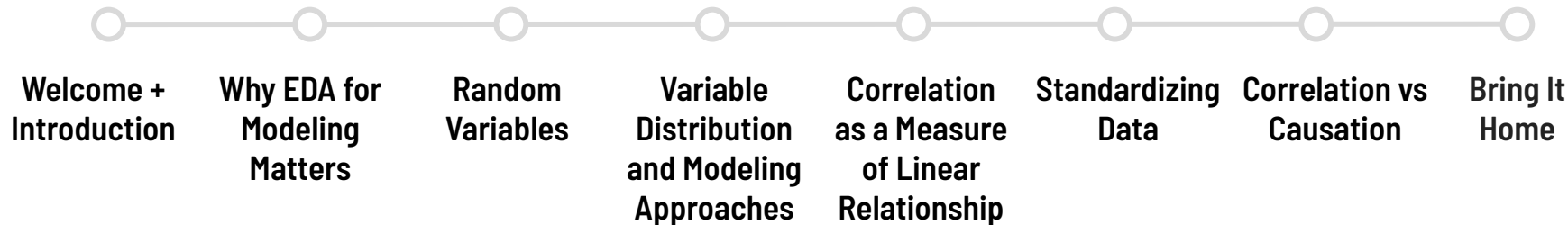
5

Explain the motivation behind and methods for standardizing data.

6

Understand that correlation itself does not imply causation.

Lesson roadmap



Why EDA for Modeling Matters

Why explore data, before modeling?

Data exploration is the foundation of effective machine learning.

The 'garbage in, garbage out' principle applies to all modeling.

Understanding your data prevents costly mistakes, and wasted resources.

EDA reveals insights that inform both model selection, and feature engineering.

Key Goals of EDA for Modeling

1. Understand data structure, quality and characteristics.
2. Identify patterns, relationships and anomalies.
3. Detect data quality issues requiring cleaning.
4. Generate hypotheses about variable relationships.
5. Inform both feature selection and engineering decisions.
6. Guide our modeling approach and algorithm selection.



Data Quality Assessment

Missing Values	Extent, patterns, and mechanisms.
Outliers	Legitimate or erroneous.
Inconsistencies	Contradictory or impossible values.
Duplicate Records	Complete or partial duplicates.
Data Types	Are they appropriate for the variables?
Class Imbalances	Distribution of the target variable.

Variable Characteristics

Distribution Shape	Normal, skewed, bimodal, uniform.
Range and Boundaries	Min/max values; truncation.
Central Tendency	Mean/median/mode.
Dispersion	Variance, standard deviation, IQR.
Scale and Unit	Absolute vs relative measurements.
Cardinality	Number of unique values (for categorical variables).

Relationship Exploration

Correlations Between Features	Strength and direction.
Correlations with Target	Potential predictive power.
Multicollinearity	Redundant information.
Interactions	Variables that work together
Hierarchical Relationships	Nested or grouped data.
Temporal Patterns	Seasonality, trends, cycles.

Common EDA Techniques

Univariate Analysis	Histograms, box plots, bar charts.
Bivariate Analysis	Scatterplots, correlation matrices.
Multivariate Analysis	Heatmaps, principal component analysis (PCA), clustering.
Statistical Summaries	Descriptive statistics.
Data Visualization	Powerful tool for pattern recognition.
Domain-Specific Analysis	Incorporating field expertise.

EDA

- ☐ What are the common pitfalls of skipping proper EDA?
- ☐ How has data exploration saved you time, or improved results, in a past project?
- ☐ How might the scope of EDA differ for different types of ML problems (classification, regression clustering)?
- ☐ How might EDA approaches differ for structured vs unstructured data?

Random Variables

Random Variable: Definition

A random variable is a variable whose value is subject to chance or randomness.

More formally, it is a function that maps outcomes of a **random process** to numerical values.



Random Variable: Example

For example, a random variable might map the possible outcomes of a coin flip $\{\text{HEADS}, \text{TAILS}\}$, or the possible outcomes of the throwing of a fair die $\{1, 2, 3, 4, 5, 6\}$, to values that sum to 1.

In data science, random variables are variables whose values we observe but cannot perfectly predict.



Types of Random Variables

Discrete

Take on distinct, countable values.
E.g., number of customers, count of defects, binary outcomes (0/1).
Described by probability mass functions (PMFs).

Continuous

Take on any value within a range.
E.g., height, weight, time, temperature. Described by probability density functions (PDFs).

Properties of Random Variables

1

Expected Value (mean)

Central value of
the distribution.

2

Variance

Spread or
dispersion
around the
mean.

3

Standard Deviation

Square root of
the variance, in
original units.

4

Quantiles

Distribution
divided into
equal portions
(e.g., median =
50th percentile).

From Theory to Practice

In data sets, we observe **realisations** of random variables.

Each observation (row) represents an outcome of the random process in question!

Columns represent different random variables we're measuring.

Statistical inference: using sample data to make conclusions about the broader population.

Random Variables in Machine Learning

Features (independent) variables: input random variables we use for prediction.

Target (dependent) variable: output random variable we aim to predict.

ML models estimate $P(Y|X)$: the probability distribution of the target (Y) given the features (X).

Goal: understand relationships between random variables.

Exploring Random Variables

Let's see it for ourselves!

- Open **02-eda-for-modeling.ipynb** parts 1, 2, and 3



Variable Distribution and Modeling Approaches

Why Distributions Matter in Modeling

Distribution shapes inform appropriate model selection.

Many models make assumptions about underlying distributions.

Transformations may be needed for optimal model performance.

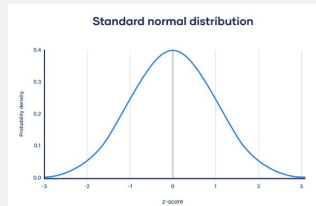
Distributions help identify outliers and anomalies.

Understanding distributions improves feature engineering.

Common Distributions in Data

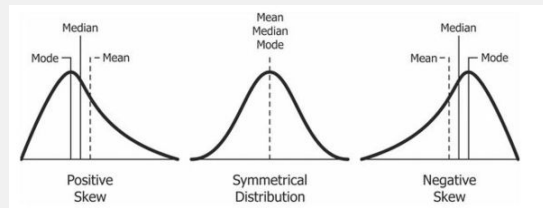
Normal (Gaussian): symmetric bell curve.

- a. Common natural phenomena caused by multiple different variables (height).
- b. Assumptions of many statistical tests and models.



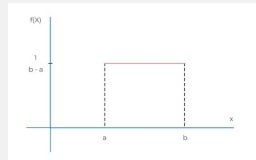
Skewed distributions: asymmetric with long tails

- a. Right-skewed: Mean > Median (income, prices, counts)
- b. Left-skewed: Mean < Median (exam scores with ceiling effects)



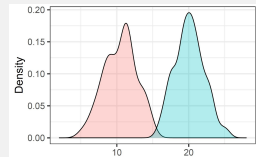
Uniform distribution: equal probability across range

- a. Random numbers, simulation inputs



Bimodal/multimodal: multiple peaks

- a. Mixed populations, day/night measurements



Distributions and Linear Models

Linear models (linear regression, logistic regression) often assume:

- a. Normal distribution of residuals
- b. Homoscedasticity (constant variance)
- c. Linear relationship between variables

Impact of violations:

- a. Biased coefficient estimates
- b. Unreliable confidence intervals
- c. Poor predictive performance

Solutions:

- a. Variable transformations (log, square root, Box-Cox)
- b. Different model families (GLMs, non-parametric models)

Distributions and Tree-based models

Tree models (decision trees, random forests, gradient boosting):

- a. Less sensitive to distribution shape
- b. No explicit distributional assumptions
- c. Can handle non-linear relationships naturally

Considerations:

- a. Extreme skewness can still affect performance
- b. Outliers may impact splitting decisions
- c. May benefit from balanced class distributions

Distributions

Scenario: Website Traffic Data

Distribution: Daily visits highly skewed (many low-traffic days, few high-traffic spikes).

- ☐ How might this distribution affect our prediction model?
- ☐ What transformations might be appropriate?

Distributions

Scenario: Customer lifetime value

Distribution: Bimodal distribution (casual vs loyal customers)?

- What modeling approach would you recommend?
- How might you handle this in feature engineering?

Distributions

Scenario: Manufacturing defect rates

Distribution:

- Highly left-skewed
- Zero-inflated data

→ What modeling considerations are important?

Correlation as a Measure of Linear Relationship

Correlation Fundamentals

Measures the direction and magnitude of a linear relationship between two variables.

Pearson correlation r ranges from -1 to $+1$; where $+1$ is perfect positive correlation, 0 is no correlation, -1 is perfect negative correlation.

Only captures linear relationships.

Interpreting Correlation Values

Strong positive (0.7-1): as X increases, Y tends to increase substantially.

Moderate positive (0.3-0.7): as X increases, Y tends to increase moderately.

Weak positive (0-0.3): as X increases, Y tends to increase slightly.

No correlation (0): no linear relationship.

Weak to strong negative (-1.0 to 0): as X increases Y tends to decrease.

Correlation Formula (for the technically minded)

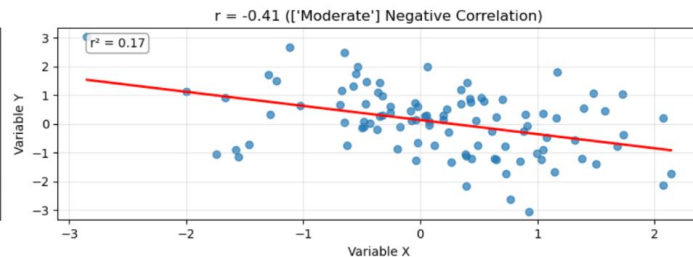
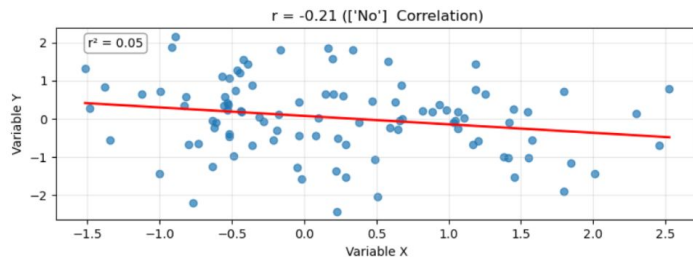
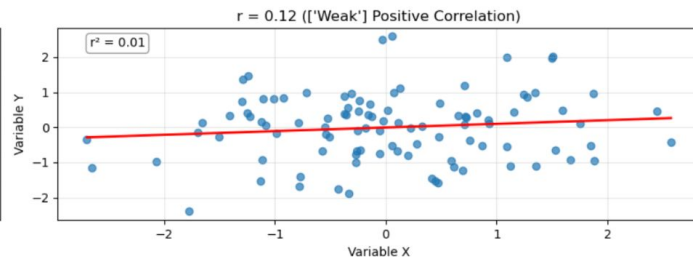
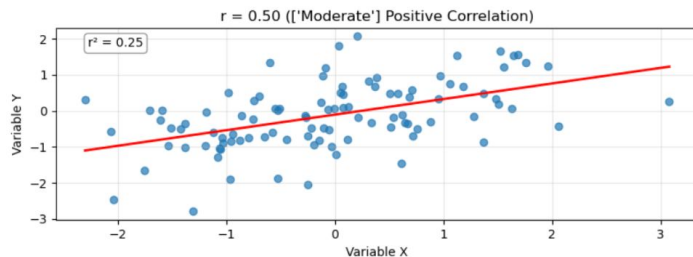
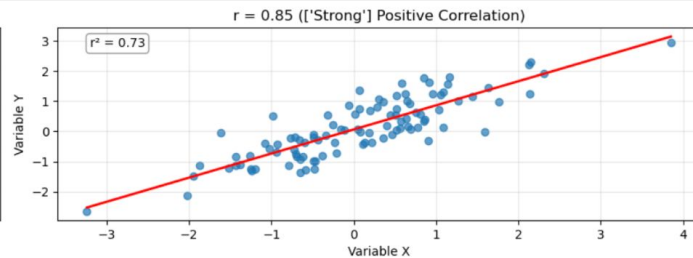
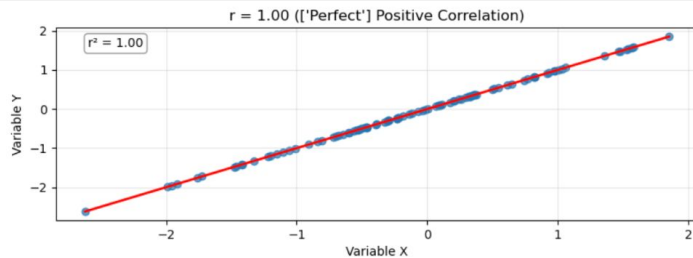
Pearson correlation coefficient formula:

$$r = \frac{n \sum x_i y_i - (\sum x_i)(\sum y_i)}{\sqrt{\left[n \sum x_i^2 - (\sum x_i)^2 \right] \left[n \sum y_i^2 - (\sum y_i)^2 \right]}} \quad \text{or} \quad r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

Measures how much two variables X, Y move together relative to their standard deviations.

Standardized covariance: the covariance divided by the product of standard deviations.

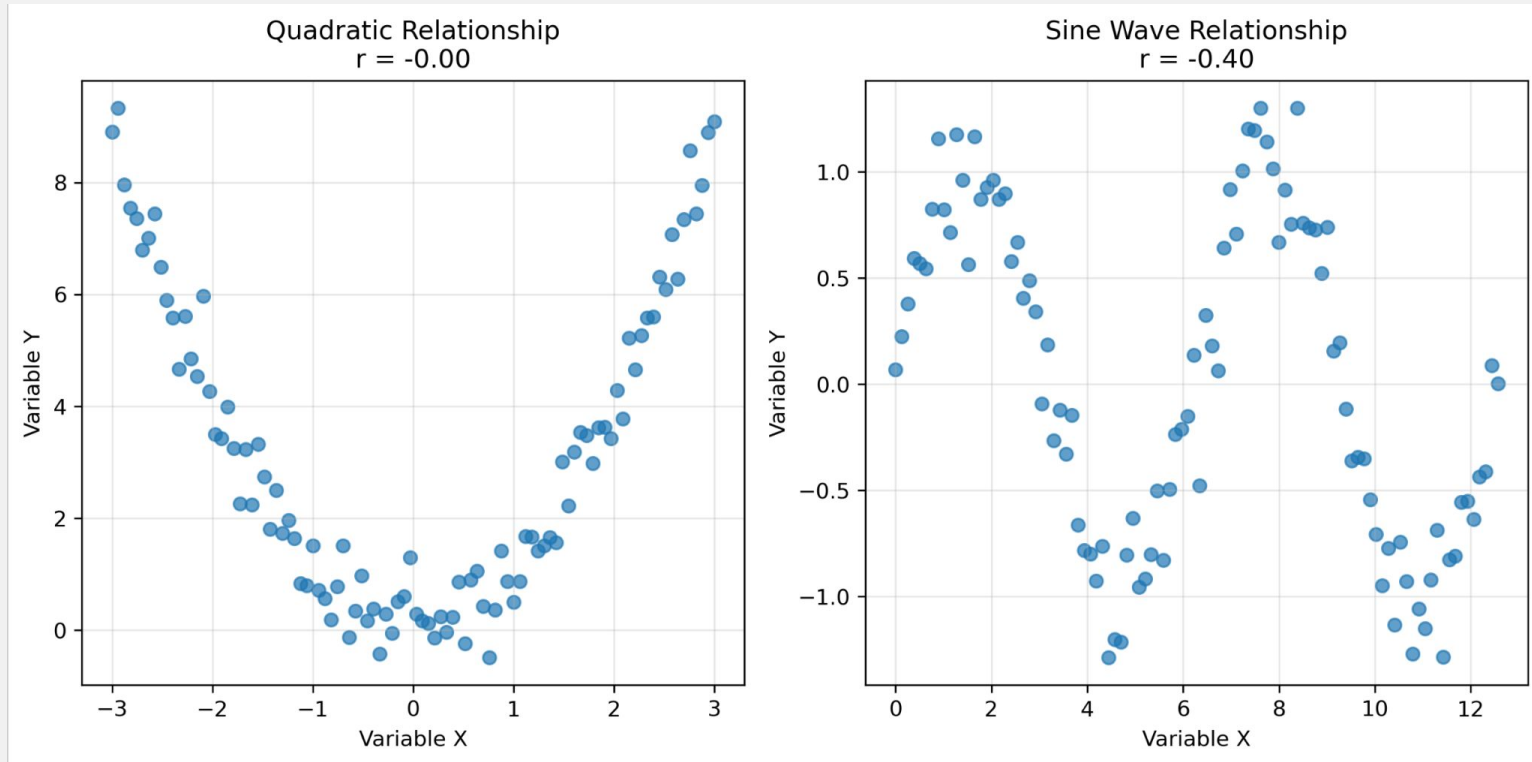
Visual Representation of Correlation



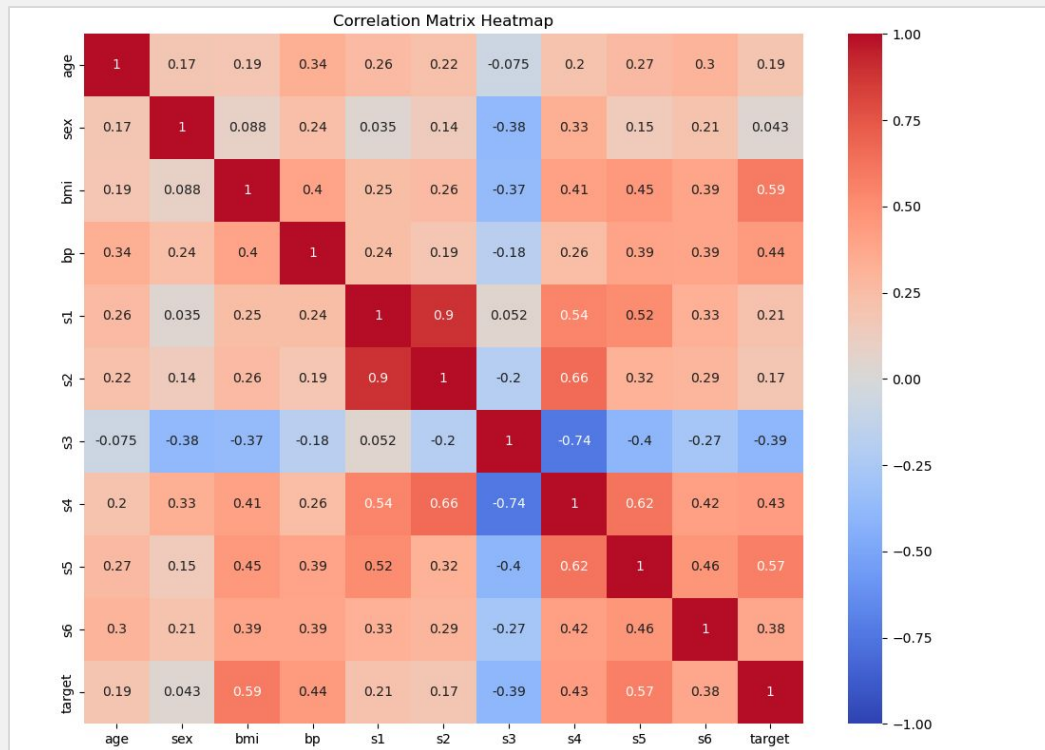
$r = -0.72$ (['Strong'] Negative Correlation)

$r = -1.00$ (['Perfect'] Negative Correlation)

Nonlinear Relationships



Correlation Heatmap



Alternative Correlation Measures

Spearman rank correlation: based on ranks, captures monotonic relationships.

Kendall's tau: another rank-based measure; but robust to outliers.

Point-Biserial correlation: between continuous and binary variables.

Distance correlation: detects nonlinear dependencies.

Mutual information: from information theory. Can represent almost any relationship.

Correlation

Let's see it for ourselves!

- Open **02-eda-for-modeling.ipynb** parts 4 and 5



Correlation:

- ☐ When might strong correlation be particularly useful for feature selection?
- ☐ What are the dangers of removing features, based solely on correlation?
- ☐ How does correlation influence multicollinearity in regression models?
- ☐ When is Spearman's *rank* more appropriate than Pearson's correlation coefficient?
- ☐ How does the notion of correlation relate to Principal Component Analysis (PCA)?

Standardizing Data

Why standardize data at all?

Features with larger scales can dominate models.

Many algorithms require standardized input:

- a. k-means clustering
- b. Principal Component Analysis (PCA)
- c. Neural networks
- d. Regularised regression models (Lasso, Ridge)
- e. Support Vector Machines

Standardization improves convergence speed.

Makes coefficient magnitudes comparable.

Necessary for proper distance-based calculations.

Common Standardization Methods

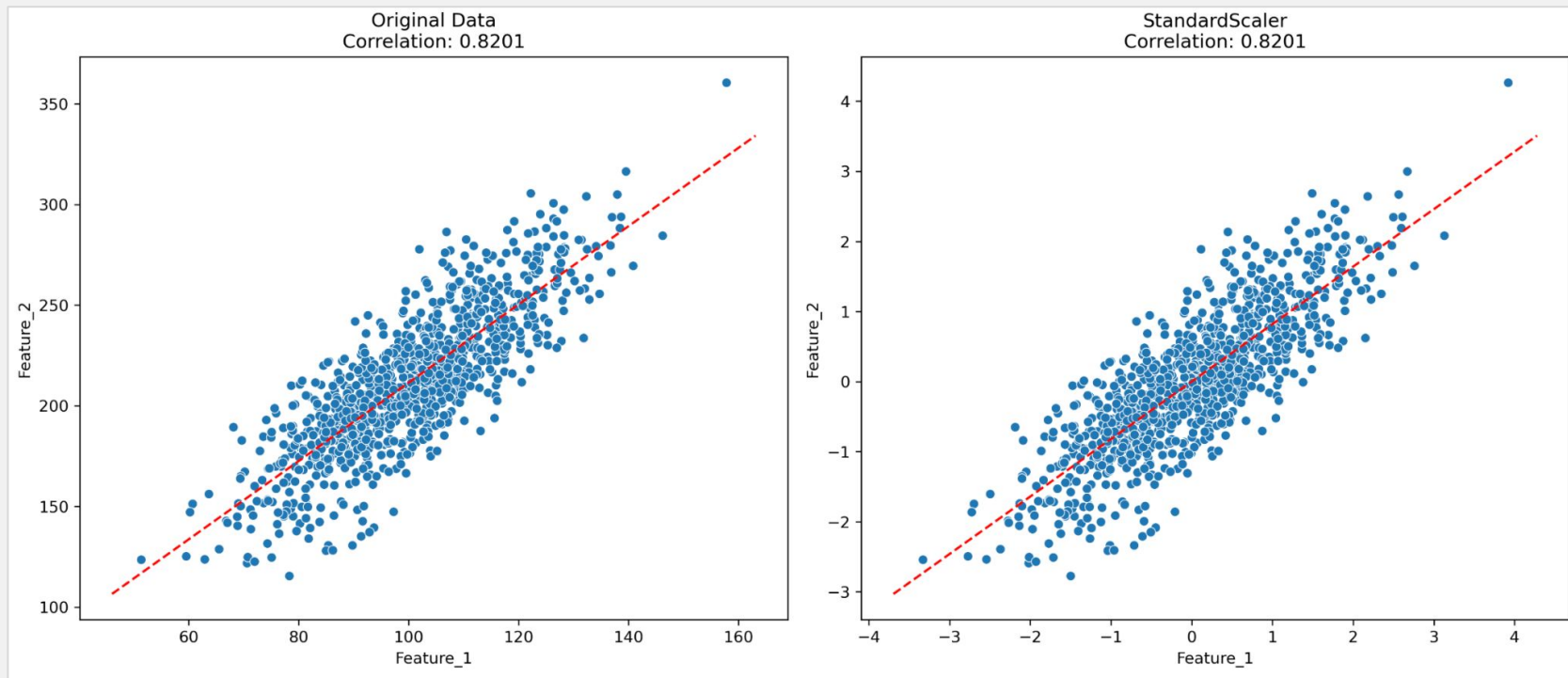
Z-score Standardization (standard scaling):

$$z_i = \frac{x_i - \mu}{\sigma}$$

Result: $\mu = 0, \sigma = 1$

Best for approximately normally distributed data, when outliers have already been addressed, and when the algorithm assumes a normal distribution.

Z-Score Standardization



Common Standardization Methods

Min-Max Scaling (Normalisation):

$$z = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

Result: Range [0, 1]

Preserves distribution shape, but sensitive to outliers. Best for when you need a bounded [0,1] interval, and useful for image data and algorithms that need positive values.

Common Standardization Methods

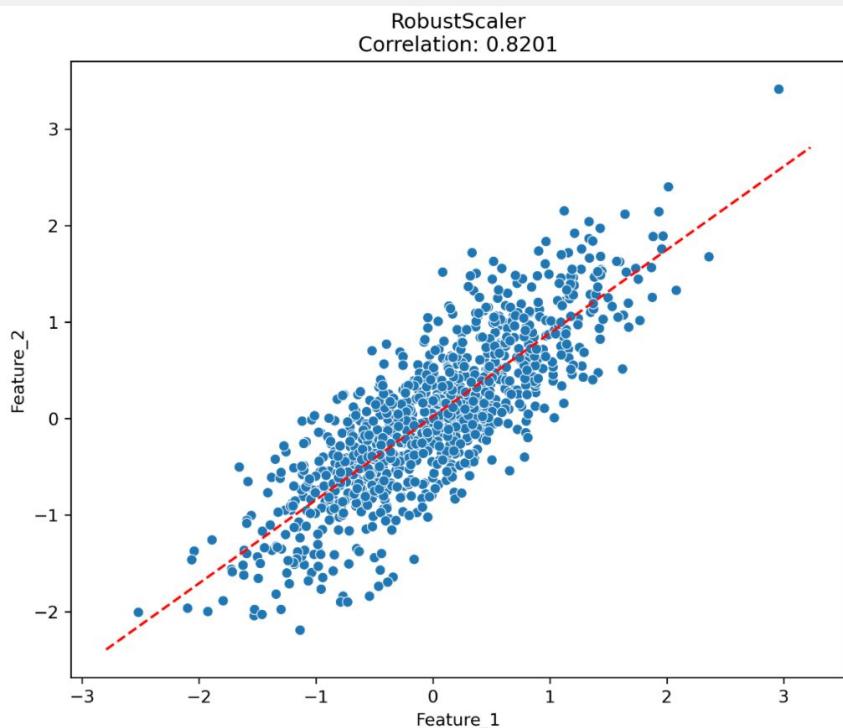
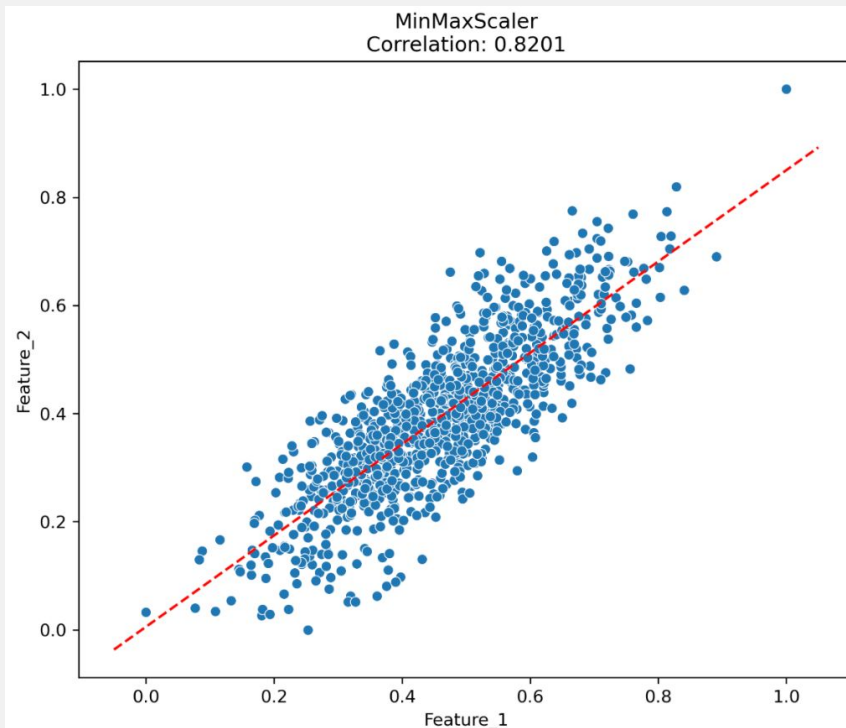
Robust scaling:

$$x' = \frac{x - \text{median}(x)}{\text{IQR}(x)}$$

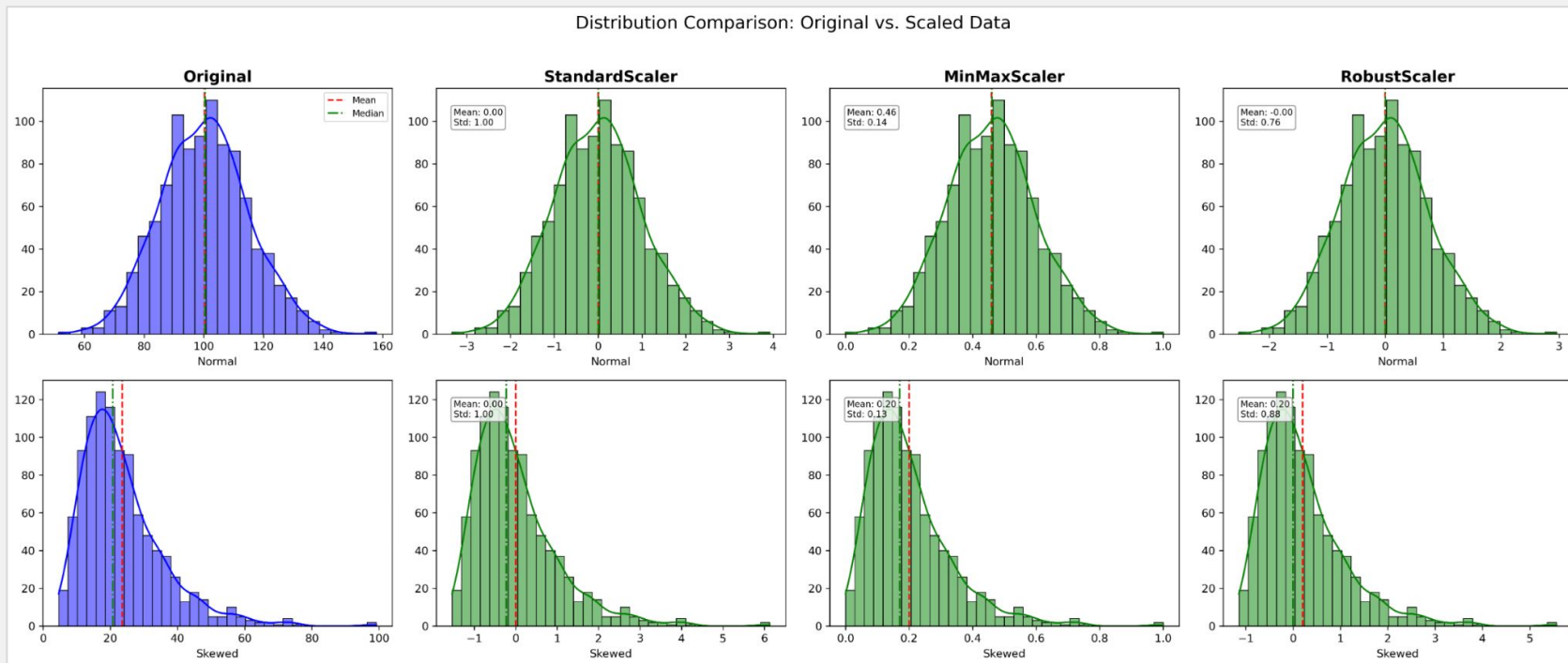
Result: Robust to outliers

Based on percentiles rather than mean/std. Useful for when data contains outliers, when the median is a better central measure than mean, and for skewed distributions.

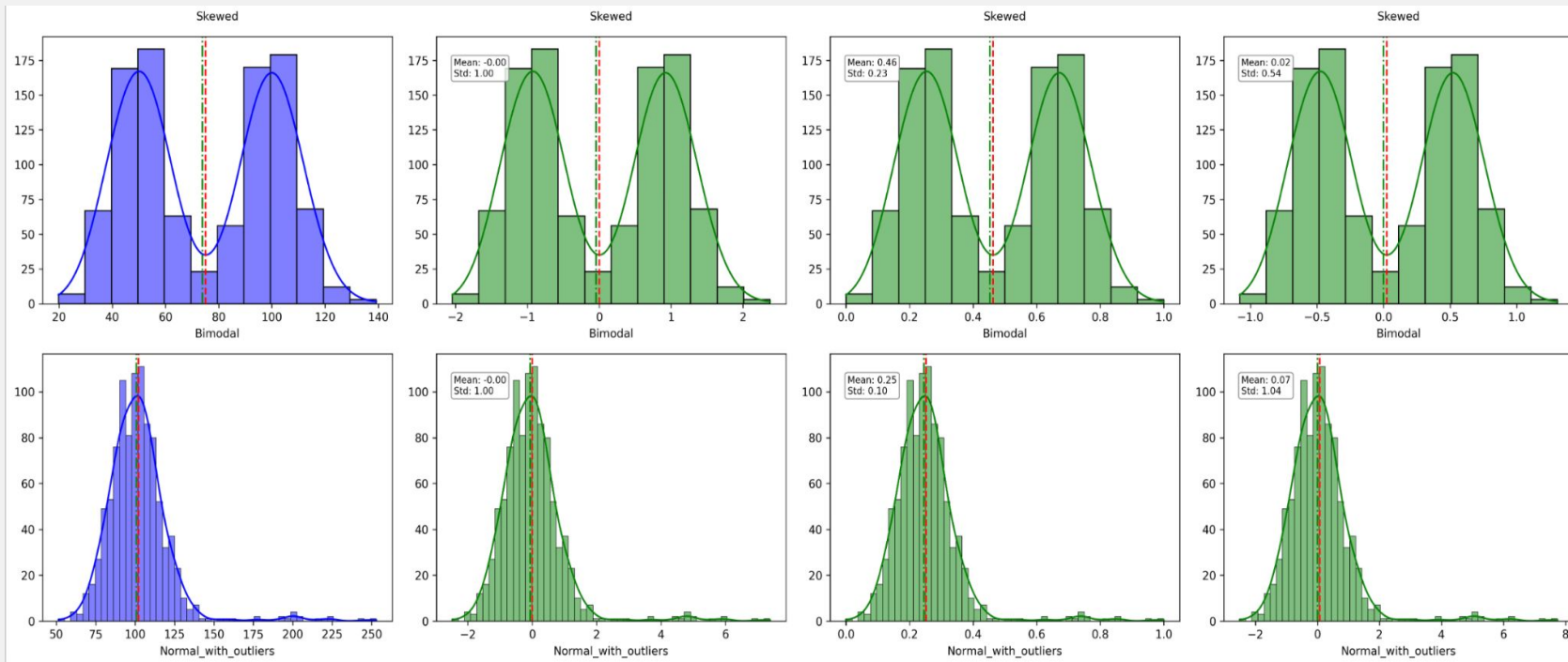
Common Standardization Methods



Common Standardization Methods



Common Standardization Methods



Standardizing Data

Let's see it for ourselves!

- Open **02-eda-for-modeling.ipynb** part 6



Correlation Vs. Causation

The Classic Warning

Correlation does not imply causation (!)

Causation is one of the most important concepts in data science.

Mistaking correlation for causation can lead to radically flawed decisions and inferences.

Finding a correlation is just the beginning of an investigation into the underlying causal structure.



What is causation?

X causes Y if a change in the value of X directly influences the value of Y.

One can establish that X causes Y only if one first verifies a statistical association, the temporal precedence of X to Y, no alternate explanations of the value of Y changing subsequent to the value of X changing, and a mechanism of action.

Even this is a ***necessary***, not ***sufficient***, condition for establishing causation (hence the 'only if', not 'if and only if', in the above).

Thus, causation is significantly harder to establish than correlation!

Common Patterns Leading to Confusion

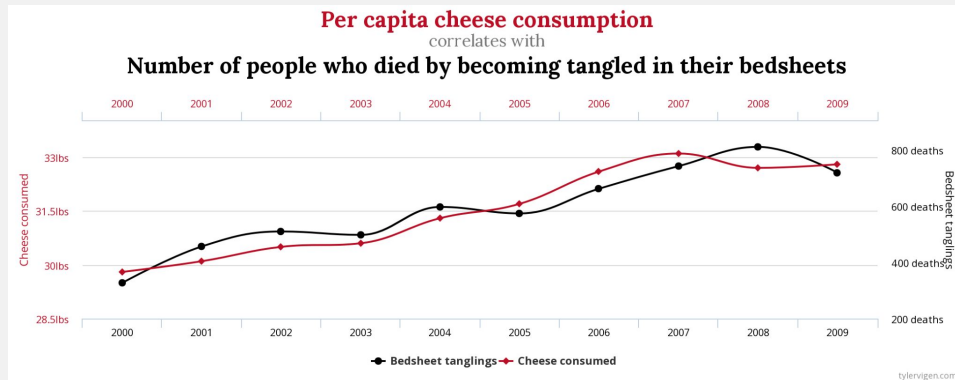
REVERSE CAUSALITY	COMMON CAUSE (CONFOUNDING)	COINCIDENTAL CORRELATION	SELECTION BIAS
The dependent variable Y causes the independent variable X, not the other way around (for instance: let X = crime cases, and Y = police officer presence).	Some third variable Z causes both the independent variable and the dependent one (e.g: X = ice cream sales, Y = shark attacks, Z = temperature).	Random chance or limited data. E.g: spurious correlations between unrelated trends.	Correlation exists due to sampling method. E.g: survey response patterns.

Famous Spurious Correlations

Nicholas Cage movies and
swimming pool drownings.

Cheese consumption and bedsheet
tangling deaths.

Divorce rate in Maine and
margarine consumption.



How to Establish Causality

Randomised Controlled Trials (RCTs) are generally taken to be the gold standard; although philosopher [Nancy Cartwright](#) disputes this.

Natural experiments.

Quasi-experimental designs.

Causal inference methods:

- Propensity score matching
- Difference-in-differences
- Instrumental variables
- Regression discontinuity

Domain knowledge and mechanism understanding

Correlation vs Causation analysis exercise

- ☐ Break into groups of 3-4 people.
- ☐ Each group will receive a data set description and correlation finding.
- ☐ For each scenario:
 - ☐ Identify if correlation likely implies causation.
 - ☐ Brainstorm alternative explanations.
 - ☐ Suggest how you would test for causality.
 - ☐ Share findings with class.

Correlation vs Causation

Health and Fitness	Strong negative correlation (-0.75) between daily step count and blood pressure in a study of 500 adults.
Education	Strong positive correlation (0.82) between hours spent on social media and declining grades among high school students.
Business	Moderate positive correlation (0.56) between company's advertising spending and sales over the past 12 months.
Environment	Strong correlation (0.91) between ice cream sales and air conditioner usage in summer months.
Finance	Correlation of 0.65 between a stock's daily trading volume and its price volatility.

Correlation vs Causation: discussion

- ☐ When is merely establishing correlation sufficient for some purposes?
When is establishing causation necessary?
- ☐ How can you incorporate causal thinking into your data analysis process?
- ☐ What tools or methods have you used to establish causality in your work?
- ☐ What are the ethical implications of making causal claims without sufficient evidence?

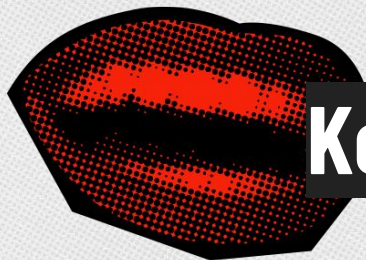
Correlation vs Causation

Let's see it for ourselves!

- Open **02-eda-for-modeling.ipynb** part 7



Bring It Home



Key takeaways



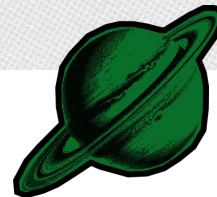
EDA is Fundamental

Data exploration is the foundation of effective machine learning.



Standardising Data Can Be Important

Some models require it.



Correlation Helps Illuminate Relationships

...but determining causation is tricky.

